

CS6600: Computer Architecture

Assignment #1

Cache Simulator

Name: **Dinesh Pitchika**

Roll No: **EE24M102**

1 Introduction

This project involved the development of a generic cache and memory hierarchy simulator, named CacheSimulator(in C++), designed to evaluate and compare various memory hierarchy configurations. By leveraging a subset of the SPEC-2000 benchmark suite, the simulator provides detailed insights into the performance, area, and energy consumption of different cache architectures. Key parameters such as like **Miss Rate**, **Average Access Time(AAT)**, **Energy Delay Product(EDP)** are analyzed and how they vary across different Cache parameters like **Size**, **Blocksize**, **Associativity** and **trade offs** is explored.

2 Experiments

In this section, the address trace provided in the *gcc.trace.txt* file was run with different memory hierarchy and with cache parameters. Variations in metrics like Miss rate, Average Access Time, EDP etc. are analyzed and trade offs among cache parameters L1 Size, Blocksize, Associativity to get optimal performance is explored.

2.1 L1 Cache Investigation: SIZE and ASSOC

2.1.1 Plot #1

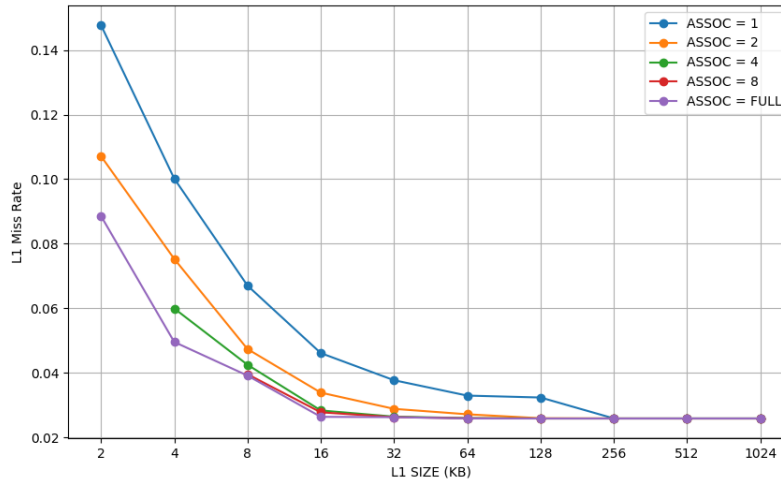


Figure 1: Miss Rate vs L1 Size for various values of L1 Assoc

In the above plots , Miss rate is being observed for different L1 sizes for a given associativity. From the graphs we can observe that the miss rate is decreasing as the cache size increases, for a given set associativity. This is because a larger cache can store more data, reducing the likelihood of cache misses.

We can also observe that miss rate is decreases as set associativity increases for a given size. This is because higher associativity reduces the likelihood of conflict misses, where multiple blocks compete for the same cache set.

For all curves, the miss rate doesn't go below **0.025**. This indicates that no matter how much the cache is, there is a minimum **compulsory miss rate of 0.025**.

For a given Set Associativity, compulsory misses and capacity misses are same for a fully associative cache of the same size, **the distance between the curves for a given associativity to that of a fully associative cache will give us the conflict misses** for a given size(which is so large).

To estimate conflict miss rate, We know that compulsory miss rate = 0.025.

Capacity Miss rate = Capacity Miss rate for FA Cache of same size = Total Miss Rate of FA - Compulsory Miss rate

Conflict Misses = Total Miss rate - Compulsory Miss rate - Capacity Miss rate = Total Miss rate - Total Miss rate of FA Cache

ASSOC	Total Miss rate	Compulsory	Capacity	Conflict
1	0.067	0.025	0.014	0.028
2	0.047	0.025	0.014	0.008
4	0.042	0.025	0.014	0.003
8	0.039	0.025	0.014	0.000
Full	0.039	0.025	0.014	0.000

Table 1: Conflict miss rate for each Associativity (for size = 8KB)

2.1.2 Plot #2

From an Average Access Time(AAT) point of view, we see that of a given set associativity, AAT reaches a minimum for a certain size and then it increases even for a bigger cache may be due to increase in hit latency for a bigger memory.

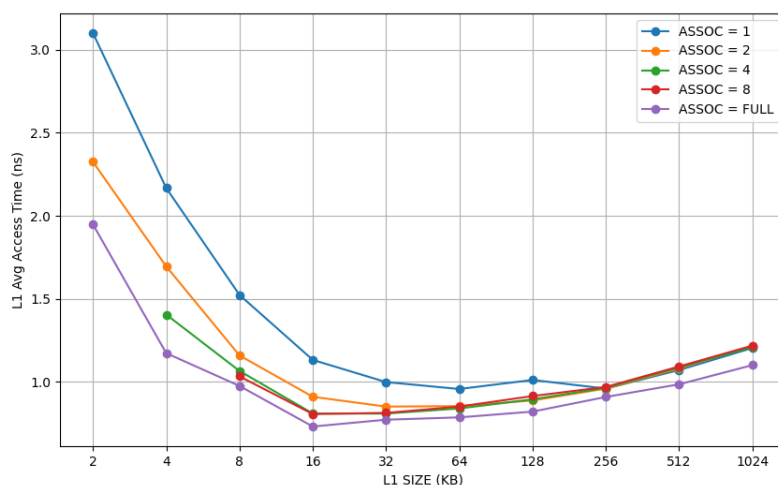


Figure 2: AAT vs L1 Size for various values of L1 Assoc

From the curves, **Fully Associative cache gives us least AAT for a given size, and CACHE SIZE = 16KB is where AAT is the least.**

2.1.3 Plot #3

From the curves, we can see that overall AAT has reduced with the addition of L2 Cache(in comparison to plot 2 latency values) and **AAT is least for Fully Associative Cache of size 16KB.**

For FA Cache of Size 16 KB,

AAT is almost the same(0.73 ns) with and without L2 Cache.

EDP is slightly higher with an L2 Cache and obviously is significantly higher with an L2 Cache.

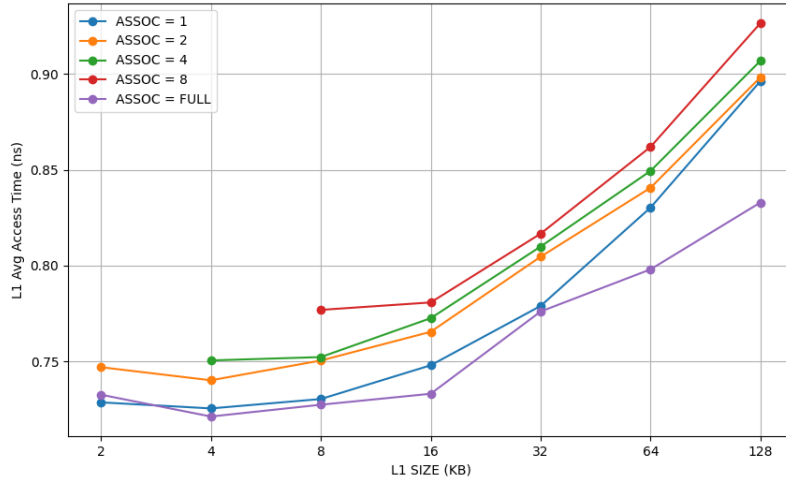


Figure 3: Average Access Time for L1 with L2 Cache

2.2 L1 Cache Investigation: SIZE and BLOCKSIZE

2.2.1 Plot #4

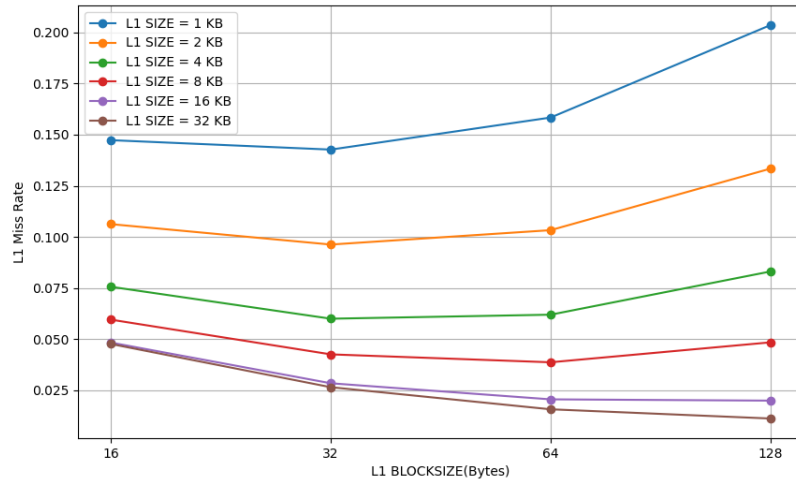


Figure 4: Miss rate vs Blocksize for various L1 Sizes

From the curves we can see that for a given size miss rate reduces only upto a certain block-size and start increasing again indicating cache pollution which invalid number of invalid bytes are being brought into the cache.

For a 4 way set associative cache of size 8KB we can see that **blocksize of 64 bytes would give us the least miss rate.**

2.3 L1 + L2 Co-optimization

2.3.1 Plot #5

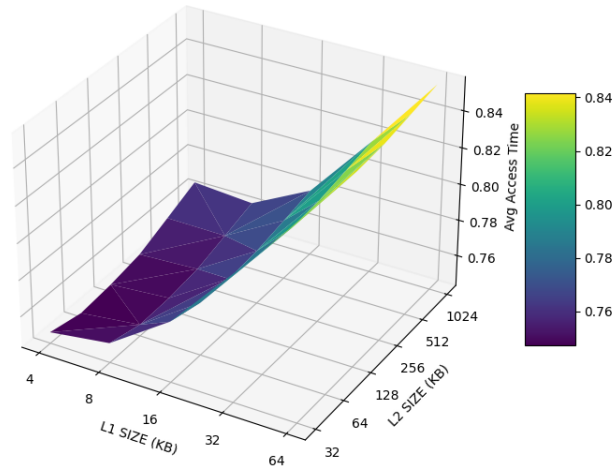


Figure 5: AAT for different L1 and L2 Sizes

From the graph, L1 Cache of 4KB and L2 Cache of 32 KB give us the best AAT.

From Plot 1 we can see that a Fully Associative L1 Cache of size 64KB gives us a better AAT.

2.3.2 Plot #6

Out of all configurations **L1 cache of 8KB along with L2 Cache of 64 KB** is found to be the one with the least EDP.

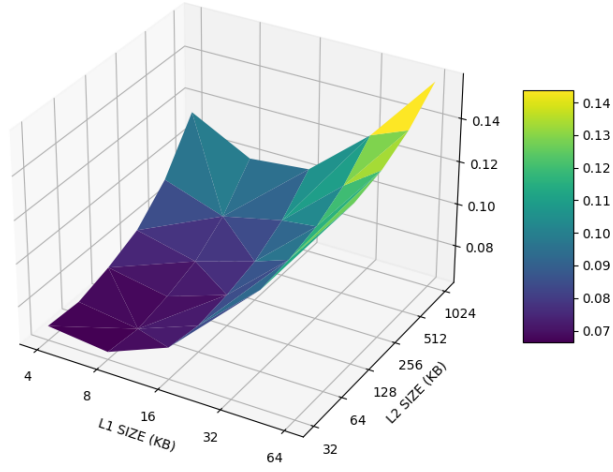


Figure 6: EDP for different L1 and L2 Sizes

2.4 Victim Cache Investigation

2.4.1 Plot #7

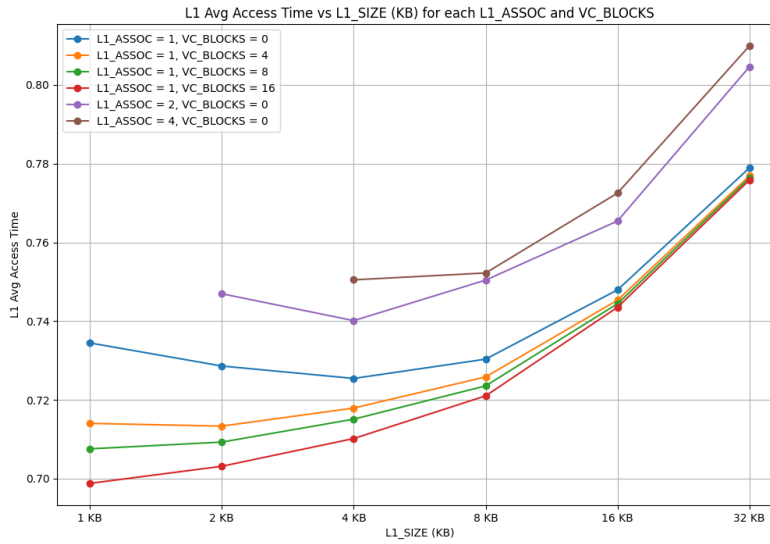


Figure 7: Miss rate for L1 with Victim Cache

From the curves, we can clearly see that direct mapped L1 with VC has been AAT than a 2 way set associative cache without VC.

Out of the curves we can observe that Direct Mapped L1 cache of size 1KB with 16 VC blocks gives us the best AAT.

From Plot 1, we can see that a Fully Associative L1 Cache of size 16KB has slightly higher AAT(within 5%) but has a significantly lesser area.